Math 118 Test 1 Knowledge Check List February 7, 2020

Here's what you need to know for test 1. The executive summary is the following: everything up to and including Lecture 12, with the exception of a few things (indicated clearly in the list below) that we did not cover thoroughly in class. I encourage you to use this list as a checklist; for each item listed below ask yourself if you understand what it means, and refer back to the lecture notes as necessary.

1. Matrices, Vectors and Norms

- Vector Norms:
  - Definitions.
  - Computing p-norms (note these are also called ℓ_p norms).
  - Cauchy-Schwarz inequality.
- Matrix norms:
  - Notion of induced matrix norm.
  - Computing $\|A\|_1, \|A\|_2$ and $\|A\|_\infty$ as well as the Frobenius norm; $\|A\|_F$.
- Matrix Multiplication:
  - Inner and outer product interpretation. These were covered in HW3.
  - Rank, and its relation to the outer product interpretation of matrix multiplication.
- Linear Systems:
  - Solving Linear Systems using the LU decomposition.
  - Finding the LU decomposition, using at most one pivot step.
  - Condition number and its affect on the solution to linear systems.
- Least Squares:
  - Motivation. That is, why use least squares instead of solving linear system.
  - Solving least squares using normal equations. Pros and cons of this approach.
  - Solving least squares with QR decomposition. Pros and cons of this approach.
  - Properties of orthogonal matrices. Particularly which norms are unitarily invariant.
  - Perturbation analysis for least squares.
- The Singular Value Decomposition:
  - Definitions and properties of singular values and singular vectors.
  - Relation between SVD and outer product interpretation of matrix multiplication. Best rank r approximation of a matrix.
  - Using SVD for PCA. Interpreting the results.
- Tensors:
  - Definitions (*i.e.* what is a tensor, what is a mode and so on).
  - Some uses of tensors.
  - Unfolding of tensors.
  - Multiplication of tensors.
  - **Note:** I will **not** test you on the HOSVD.
- k-means:
  - When to use k-means (*i.e.* what is a clustering problem).

  - implementing k-means: either pseudocode or running an iteration or two on a toy data set.
  - The Quality score, or function that k-means implicitly seeks to minimize.
  - When not to use k-means (*i.e.* what sort of data sets it is likely to fail on).
- Non-negative matrix factorization:
  - Definition.
  - Setting this up as an optimization problem.
  - Using this to extract relevant topics from text data, segment customers etc.
- Graphs:
  - Basic definitions (*i.e.* vertices/nodes, edges, adjacency matrix etc.
  - Intuitive notion of what a cluster is.
  - Spectral Clustering:
    * Heuristic motivation for spectral clustering in terms of matrix perturbation.
    * Cut, Ratio Cut and its relation to spectral clustering.
    * Implementing spectral clustering: either pseudocode or performing it on a small data set, given the eigenvectors.
  - **Note:** I will **not** be testing you on centrality measures.
- Classifying Handwritten Digits
  - Being able to describe and/or recognize a classification problem.
  - Understanding how images can be converted to matrices.
  - the k-NN classifier:
    * Basic idea and justification.
    * Implementing this algorithm, either pseudocode or a toy example.
    * Pros and cons.
  - The SVD bases classifier:
    * Idea behind this, in particular why we might expect images in the same class to live in a low dimensional space.
    * Implementation: I won't ask you for pseudocode or to do a toy example, but make sure you understand all the steps and are ready to fill in the gaps if I give you partially completed pseudocode.
    * The definition of a projection matrix.
    * Pros and cons.
- Text pre-processing and Text retrieval:
  - I won't test you on tokenizing, stemming, lemmatizing etc. but you should at least have some idea of what sort of preprocessing needs to be done on text documents.
  - tf-idf: Know what it is, why its important and how to use it.
  - Non-negative matrix factorization to extract topics from a corpus of documents.
  - **Note:** I will **not** test you on query matching.
- Linear Programming:
  - Converting a word problem to a Linear Program.
  - Definitions: constraints, feasible region, cost function etc.
  - Solving simple LP's graphically.
  - Converting LP's to standard form using slack variables.
  - **Note:** I will **not** be examining you on the simplex method.